Day 060 | Unit 04 | Teacher plan

Numeric and Algebraic Expressions

Unit 4 Test

Standards: 6.NS.C.5, 6.NS.C.6a, 6.NS.C.6c, 6.EE.A.1, 6.EE.A.2c, 6.NS.B.4a, 6.NS.B.4b, 6.NS.B.4c, 6.EE.A.2b, 6.EE.A.2a, 6.EE.B.6, 6.EE.A.3, 6.EE.A.4

Learning target: I can choose a model, solve accurately, explain my reasoning, and check the result.

enVision Topic 3: Numeric and Algebraic Expressions | Spectrum companion Chapter 5

Assessment routine

5 minutes: explain response types without solving any item. 35 minutes: independent assessment. 10 minutes: collect work and note misconceptions for the next lesson. Keep the answer key and worked-example help unavailable during assessment.

Review allocation

Questions 1–4 are previous-unit review (20%). Current-unit questions increase in demand after the review. Unit 1 uses Grade 5 readiness skills because no earlier course unit exists.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 1, 8, 14, 16, 17, 18, 20, 22

Day 060 | Unit 04 | Unit Test

Numeric and Algebraic Expressions

Unit 4 Test

Standards: 6.NS.C.5, 6.NS.C.6a, 6.NS.C.6c, 6.EE.A.1, 6.EE.A.2c, 6.NS.B.4a, 6.NS.B.4b, 6.NS.B.4c, 6.EE.A.2b, 6.EE.A.2a, 6.EE.B.6, 6.EE.A.3, 6.EE.A.4

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enVision Topic 3: Numeric and Algebraic Expressions | Spectrum companion Chapter 5

Directions

The first 4 questions (20%) review Unit 3. The remaining questions assess this unit. Work in order and show your reasoning. Each whole item receives equal credit; a multi-select item requires every correct selection and no incorrect selection.

20 questions | 100 points: 5 per question

01 / Previous unit review | Constructed response

A marker is 28 m above sea level. Write its signed elevation and explain what elevation 0 means.

02 / Previous unit review | Constructed response

Find the opposite of -8. Show both values on a number line.

03 / Previous unit review | Constructed response

Draw a horizontal or vertical number line. Plot and label -8, 0, and 7.

04 / Previous unit review | Constructed response

Draw a horizontal or vertical number line. Plot and label -5, 0, and 3.

05 / Constructed response

Write 4^2 as repeated multiplication and evaluate.

06 / Constructed response

Evaluate 5 + 4 × 4. Show the intermediate steps.

07 / Constructed response

Find the GCF of 10 and 15. Show factor pairs or prime factors.

08 / Constructed response

Find the LCM of 8 and 9 by listing multiples.

09 / Constructed response

Find the GCF of 24 and 36. Show factor pairs or prime factors.

10 / Constructed response

Factor 28 + 42 using its GCF. Make the inside numbers relatively prime.

11 / Constructed response

For (5x + 7) ÷ 2, name the outer operation, the dividend as a single entity, and the divisor. Identify the sum inside the parentheses.

12 / Expression entry and reasoning

A student buys x markers at $4 each and y folders at $3 each, then uses one $2 coupon. Part A: enter an expression for the total cost. Part B: explain why the coupon is subtracted only once.

13 / Numeric entry and reasoning

Let x = 1 and y = 3. Evaluate $4x + y^2$. Part A: enter the value. Part B: show substitution and the order of operations.

14 / Multiple select

Select all expressions equivalent to 7(x + 6).
A. 42 + 7x
B. 13x
C. 7x + 6
D. 7x + 42

15 / Multiple select

Select all expressions equivalent to 6(x + 5).
A. 11x
B. 6x + 5
C. 30 + 6x
D. 6x + 30

16 / Multiple select

Select all expressions equivalent to 2x + 5x + 4.
A. 7x + 4
B. (2 + 5)x + 4
C. 7x + 5
D. 11x

17 / Multiple select

Select all expressions equivalent to 4(x + 6).
A. 10x
B. 4x + 6
C. 4x + 24
D. 24 + 4x

18 / Numeric entry and reasoning

Let x = 3/2 and y = 2. Evaluate 5x + y^2. Part A: enter the value. Part B: show substitution and the order of operations.

19 / Multiple select

Select all expressions equal to 6^3.
A. 6 × 3
B. 6 × 6 × 6
C. 217
D. 216

20 / Numeric entry and error analysis

Evaluate 3^2 + 3(6 - 3). Part A: enter the value. Part B: a student begins by adding 3^2 and 3; explain why that changes the expression.

Day 060 | Unit 04 | Answer key

Numeric and Algebraic Expressions

Unit 4 Test

Standards: 6.NS.C.5, 6.NS.C.6a, 6.NS.C.6c, 6.EE.A.1, 6.EE.A.2c, 6.NS.B.4a, 6.NS.B.4b, 6.NS.B.4c, 6.EE.A.2b, 6.EE.A.2a, 6.EE.B.6, 6.EE.A.3, 6.EE.A.4

Learning target: I can choose a model, solve accurately, explain my reasoning, and check the result.

1. +28 m; 0 means at sea level

Sea level is the reference. Below is negative and above is positive.

6.NS.C.5 | Constructed response

2. 8

-8 and 8 are each 8 units from 0 on opposite sides.

6.NS.C.6a | Constructed response

3. -8 < 0 < 7; unit steps

Negative -8 is left of or below 0; 7 is right of or above 0. Use equal intervals.

6.NS.C.6c | Constructed response

4. -5 < 0 < 3; unit steps

Negative -5 is left of or below 0; 3 is right of or above 0. Use equal intervals.

6.NS.C.6c | Constructed response

5. 4 × 4 = 16

The exponent counts 2 factors of 4; multiply all factors.

6.EE.A.1 | Constructed response

6. 21

Use parentheses and exponents first, then × and ÷ left to right, then + and -. Result: 21.

6.EE.A.1, 6.EE.A.2c | Constructed response

7. 5

Common factors: 1, 5; greatest is 5.

6.NS.B.4a | Constructed response

8. 72

The first positive multiple shared by 8 and 9 is 72.

6.NS.B.4b | Constructed response

9. 12

Common factors: 1, 2, 3, 4, 6, 12; greatest is 12.

6.NS.B.4a | Constructed response

10. 14(2 + 3)

GCF = 14; 28 = 14 × 2 and 42 = 14 × 3.

6.NS.B.4c | Constructed response

11. Outer quotient; dividend (5x + 7); divisor 2; inner sum 5x + 7

The grouped sum is one complete quantity being divided. Within it, the terms are 5x and 7.

6.EE.A.2b | Constructed response

12. 4x + 3y - 2

Marker cost is 4x; folder cost is 3y. Add these subtotals, then subtract the one fixed coupon of 2. An expression does not need an equals sign.

6.EE.A.2a, 6.EE.B.6 | Expression entry and reasoning

13. 13

Substitute: 4 × (1) + 3^2. The product is 4 and the square is 9. Add to get 13.

6.EE.A.2c | Numeric entry and reasoning

14. A, D - 42 + 7x; 7x + 42

Distribute 7 to both terms to get 7x + 42. Reversing the addends gives the same value. The other expressions fail, for example at x = 0.

6.EE.A.3 | Multiple select

15. C, D - 30 + 6x; 6x + 30

Distribute 6 to both terms to get 6x + 30. Reversing the addends gives the same value. The other expressions fail, for example at x = 0.

6.EE.A.3 | Multiple select

16. A, B - 7x + 4; (2 + 5)x + 4

Combine coefficients of like x terms: 2 + 5 = 7. The constant 4 stays separate.

6.EE.A.3 | Multiple select

17. C, D - 4x + 24; 24 + 4x

Distribute 4 to both terms to get 4x + 24. Reversing the addends gives the same value. The other expressions fail, for example at x = 0.

6.EE.A.4 | Multiple select

18. 23/2

Substitute: 5 × (3/2) + 2². The product is 15/2 and the square is 4. Add to get 23/2.

6.EE.A.2c | Numeric entry and reasoning

19. B, D - 6 × 6 × 6; 216

The exponent 3 means three factors of 6; their product is 216. It does not mean 6 times 3.

6.EE.A.1 | Multiple select

20. 18

Parentheses: 6 - 3 = 3. Exponent: 3² = 9. Multiply: 3 × 3 = 9. Add: 9 + 9 = 18. The product must be completed before the final sum.

6.EE.A.1, 6.EE.A.2c | Numeric entry and error analysis